In [ ]:
```
# EXPERIMENT 5 – DATA VISUALIZATION WITH MATPLOTLIB
PIN NO.: 241UDS0110

## AIM
To visualize and analyze a dataset using line plots, scatter plots, subplots, and time-series comparisons with
Matplotlib.
```

In [ ]:
```
## Step 1: Import the required libraries & dataset
```

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt

df_241UDS0110 = pd.read_csv("Experiment_5_241UDS0110_Campus_Energy_Usage.csv")
df_241UDS0110.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 17 entries, 0 to 16
Data columns (total 13 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Year                        17 non-null     int64
 1   Campus_Production_MWh       17 non-null     float64
 2   Campus_Consumption_MWh      17 non-null     float64
 3   Grid_Imports_MWh            17 non-null     float64
 4   Electricity_Cost_INR        17 non-null     int64
 5   Fuel_Cost_USD               17 non-null     float64
 6   Inflation_Rate              17 non-null     float64
 7   Energy_Subsidy_INR          17 non-null     int64
 8   Beneficiaries_Thousands     17 non-null     int64
 9   Import_Dependency_Percent   17 non-null     float64
 10  Consumption_Growth_Percent  17 non-null     float64
 11  Cost_to_Fuel_Ratio          17 non-null     float64
 12  Subsidy_per_MWh             17 non-null     float64
dtypes: float64(9), int64(4)
memory usage: 1.9 KB
```

In [ ]:
```
## Step 2: Display the dataset
```

In [2]:
```
df_241UDS0110.head()
```

Out[2]: Columns 1–8 of 14

	Year	Campus_Production_MWh	Campus_Consumption_MWh	Grid_Imports_MWh	Electricity_Cost_INR	Fuel_Cost_USD	Inflation_Rate
0	2010	8.4	12.2	3.8	280	58.0	8.2
1	2011	8.7	13.1	4.4	300	62.5	7.6
2	2012	8.9	13.8	4.9	315	65.0	6.9
3	2013	9.3	14.6	5.2	330	68.0	7.1
4	2014	9.6	15.3	5.7	360	72.0	6.5

Columns 9–14 of 14 · first column repeated

	Energy_Subsidy_INR	Beneficiaries_Thousands	Import_Dependency_Percent	Consumption_Growth_Percent	Cost_to_Fuel_Ratio	Subsidy_per_MWh
0	12000	80	31.15	0.00	4.83	983.61
1	14500	88	33.59	7.38	4.80	1106.87
2	16000	95	35.51	5.34	4.85	1159.42
3	18500	102	35.62	5.80	4.85	1267.12
4	21000	110	37.25	4.79	5.00	1372.55

In [3]:
```
df_241UDS0110
```

Out[3]: Columns 1–8 of 14

	Year	Campus_Production_MWh	Campus_Consumption_MWh	Grid_Imports_MWh	Electricity_Cost_INR	Fuel_Cost_USD	Inflation_Rate
0	2010	8.4	12.2	3.8	280	58.0	8.2
1	2011	8.7	13.1	4.4	300	62.5	7.6
2	2012	8.9	13.8	4.9	315	65.0	6.9
3	2013	9.3	14.6	5.2	330	68.0	7.1
4	2014	9.6	15.3	5.7	360	72.0	6.5
5	2015	10.1	16.2	6.1	390	70.0	5.8
6	2016	10.5	17.0	6.5	420	55.0	4.9
7	2017	10.8	18.1	7.3	455	57.5	4.2
8	2018	11.2	19.4	8.2	480	61.0	3.9
9	2019	11.7	20.0	8.6	510	63.5	4.5
10	2020	11.5	20.8	9.3	540	50.0	6.1
11	2021	12.0	21.9	9.9	575	67.0	5.4
12	2022	12.4	22.6	10.2	610	74.0	6.0
13	2023	12.9	23.8	10.9	640	78.0	5.1
14	2024	13.1	24.5	11.4	670	73.5	4.7
15	2025	13.7	25.7	12.0	710	69.0	3.8
16	2026	14.0	26.4	12.6	745	82.0	4.1

Columns 9–14 of 14 · first column repeated

	Energy_Subsidy_INR	Beneficiaries_Thousands	Import_Dependency_Percent	Consumption_Growth_Percent	Cost_to_Fuel_Ratio	Subsidy_per_MWh
0	12000	80	31.15	0.00	4.83	983.61
1	14500	88	33.59	7.38	4.80	1106.87
2	16000	95	35.51	5.34	4.85	1159.42
3	18500	102	35.62	5.80	4.85	1267.12
4	21000	110	37.25	4.79	5.00	1372.55
5	19500	117	37.65	5.88	5.57	1203.70
6	11000	124	38.24	4.94	7.64	647.06
7	9800	133	40.33	6.47	7.91	541.44
8	13500	145	42.27	7.18	7.87	695.88
9	18000	156	43.00	3.09	8.03	900.00
10	22000	165	44.71	4.00	10.80	1057.69
11	17500	174	45.21	5.29	8.58	799.09
12	9200	183	45.13	3.20	8.24	407.08
13	10500	194	45.80	5.31	8.21	441.18
14	12500	205	46.53	2.94	9.12	510.20
15	14000	217	46.69	4.90	10.29	544.75
16	11500	228	47.73	2.72	9.09	435.61

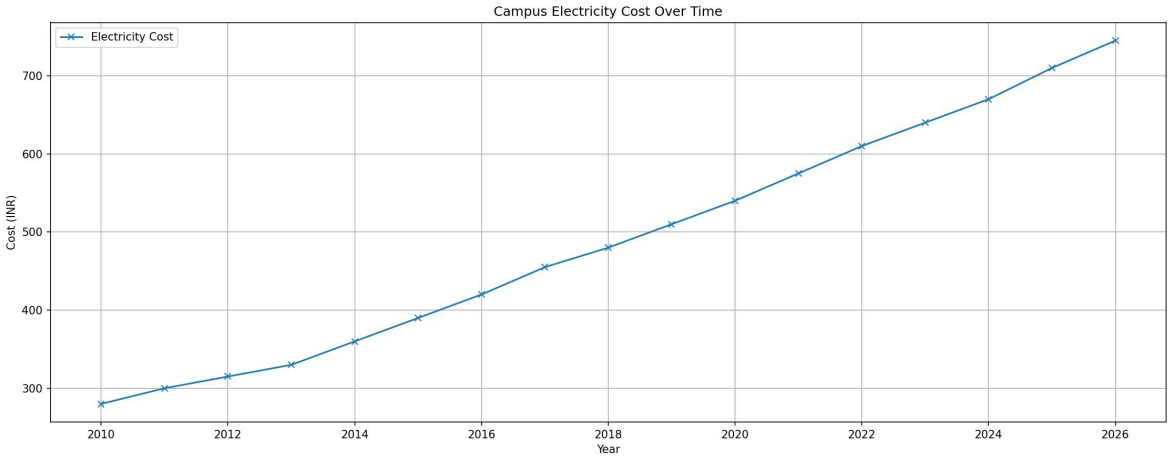
In [4]:
```
df_241UDS0110.columns
```

Out[4]:
```
Index(['Year', 'Campus_Production_MWh', 'Campus_Consumption_MWh',
       'Grid_Imports_MWh', 'Electricity_Cost_INR', 'Fuel_Cost_USD',
       'Inflation_Rate', 'Energy_Subsidy_INR', 'Beneficiaries_Thousands',
       'Import_Dependency_Percent', 'Consumption_Growth_Percent',
       'Cost_to_Fuel_Ratio', 'Subsidy_per_MWh'],
      dtype='str')
```

In [6]:
```
df_241UDS0110["Year"] = pd.to_numeric(df_241UDS0110["Year"])
```

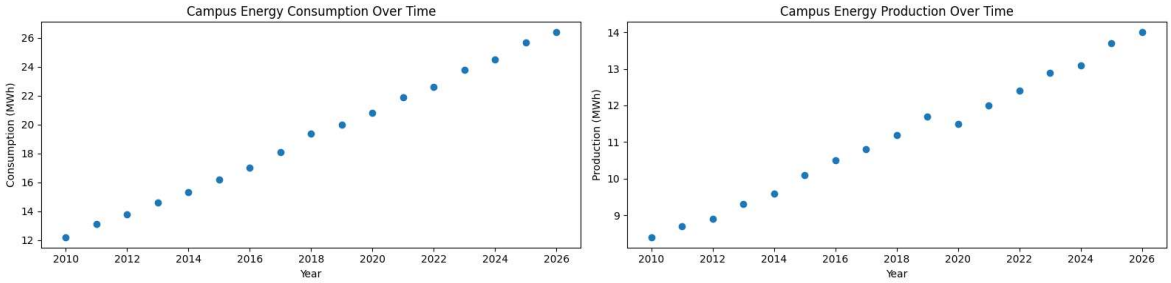
In [ ]:
```
## Step 3: Visualize Data Using Various Plots
```

In [7]:
```
plt.figure(figsize=(15, 6), dpi=150)
plt.plot(
    df_241UDS0110["Year"],
    df_241UDS0110["Electricity_Cost_INR"],
    marker="x",
    label="Electricity Cost"
)
plt.title("Campus Electricity Cost Over Time")
plt.xlabel("Year")
plt.ylabel("Cost (INR)")
plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```



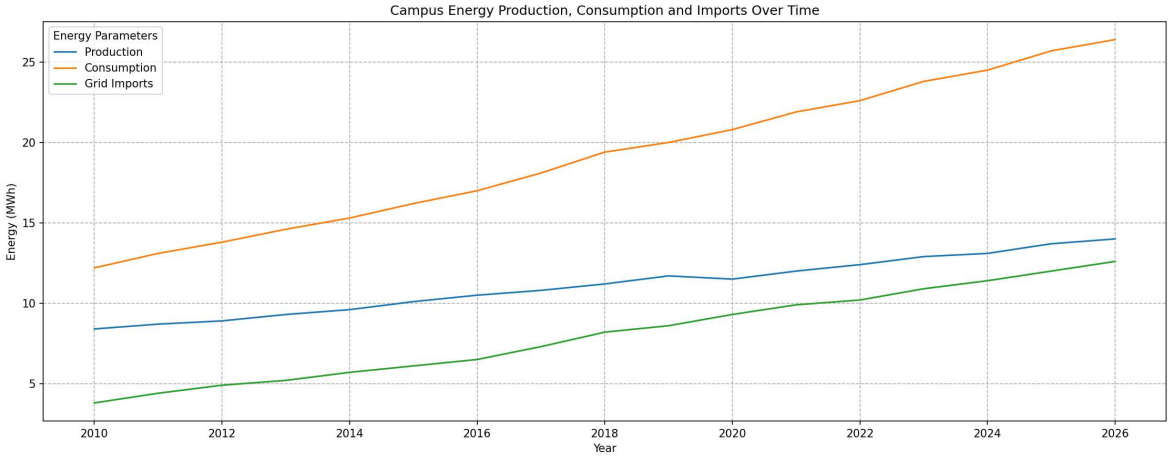
In [ ]:
```
## Step 4: Subplots
```

In [8]:
```
fig, axes = plt.subplots(1, 2, figsize=(16, 4))
axes[0].scatter(df_241UDS0110["Year"], df_241UDS0110["Campus_Consumption_MWh"])
axes[0].set_title("Campus Energy Consumption Over Time")
axes[0].set_xlabel("Year")
axes[0].set_ylabel("Consumption (MWh)")
axes[1].scatter(df_241UDS0110["Year"], df_241UDS0110["Campus_Production_MWh"])
axes[1].set_title("Campus Energy Production Over Time")
axes[1].set_xlabel("Year")
axes[1].set_ylabel("Production (MWh)")
plt.tight_layout()
plt.show()
```



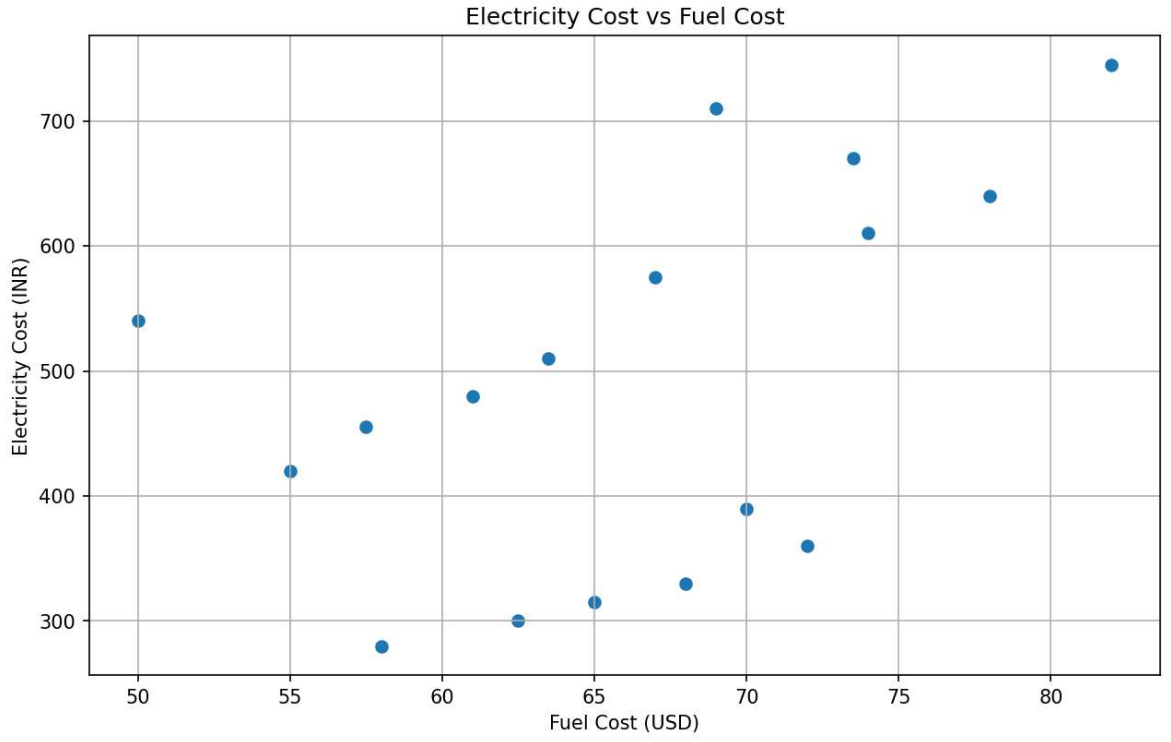
In [ ]:
```
## Step 5: Comparing Production, Consumption and Imports
```

In [9]:
```
plt.figure(figsize=(15, 6), dpi=150)
plt.plot(df_241UDS0110["Year"], df_241UDS0110["Campus_Production_MWh"], label="Production")
plt.plot(df_241UDS0110["Year"], df_241UDS0110["Campus_Consumption_MWh"], label="Consumption")
plt.plot(df_241UDS0110["Year"], df_241UDS0110["Grid_Imports_MWh"], label="Grid Imports")
plt.title("Campus Energy Production, Consumption and Imports Over Time")
plt.xlabel("Year")
plt.ylabel("Energy (MWh)")
plt.tight_layout()
plt.legend(loc="upper left", title="Energy Parameters")
plt.grid(True, linestyle="--")
plt.show()
```



In [ ]:
```
## Step 6: Electricity Cost vs Fuel Cost
```

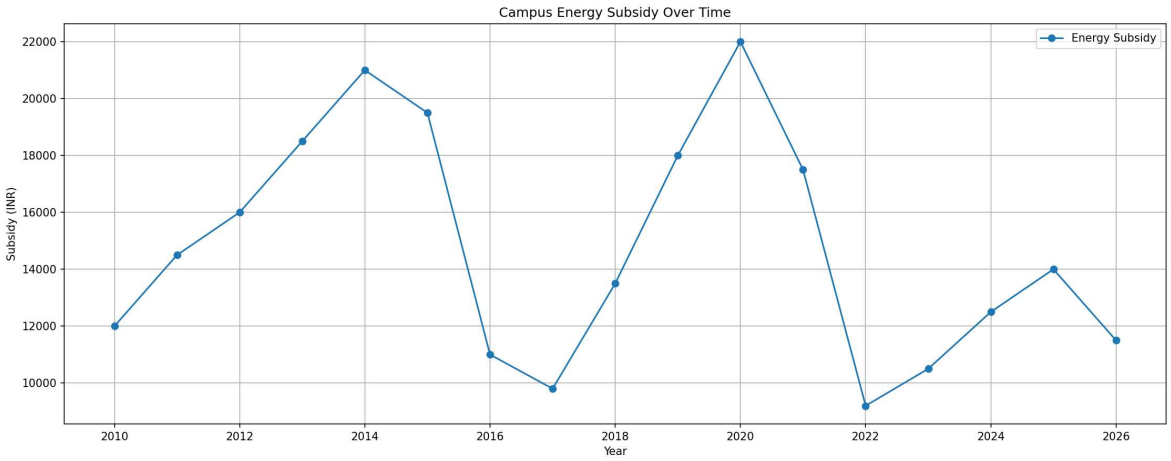
In [10]:
```
plt.figure(figsize=(10, 6), dpi=150)
plt.scatter(df_241UDS0110["Fuel_Cost_USD"], df_241UDS0110["Electricity_Cost_INR"])
plt.title("Electricity Cost vs Fuel Cost")
plt.xlabel("Fuel Cost (USD)")
plt.ylabel("Electricity Cost (INR)")
plt.grid(True)
plt.show()
```



In [ ]:
```
## Step 7: Energy Subsidy Over Time
```

In [11]:

```
plt.figure(figsize=(15, 6), dpi=150)
plt.plot(df_241UDS0110["Year"], df_241UDS0110["Energy_Subsidy_INR"], marker="o", label="Energy Subsidy")
plt.title("Campus Energy Subsidy Over Time")
plt.xlabel("Year")
plt.ylabel("Subsidy (INR)")
plt.tight_layout()
plt.legend()
plt.grid(True)
plt.show()
```



In [ ]:

```
## RESULT
Matplotlib was used to visualize campus energy costs, consumption, production, grid imports, the relationship
between electricity and fuel costs, and subsidy trends over time.
```